BTS (C) – V(S) – 11 – 038 - K

B.Tech Degree V Semester (Supplementary) Examination June 2011

ME 505 HEAT AND MASS TRANSFER

(2002 Scheme)

(Data book is permitted)

Time : 3 Hours Maximum Marks : 100

I. a) Explain the various modes of heat transfer. (10)

b) A brick wall (k=07W/mk) is 0.30m thick. If the temperature of the inner and outer surfaces are maintained at 50° and 30°C respectively, calculate the heat loss through one square metre area. Also find the temperature at an interior point of the wall 24cm distant from the outer wall. (10)

OR

II. a) Explain the significance of fin effectiveness and fin efficiency. (5)

b) What is the importance of insulated tip solution for the fins used in practice? (5)

c) An aluminium fin (k=200 W/mk) 3mm thick and 7.5cm long protrudes from a wall at 300°C. The ambient temperature is 50°C with $h=10W/m^2k$. Compute the heat loss from the fin per unit depth of material. Also calculate its efficiency and effectiveness. (10)

III. a) Distinguish between natural and forced convection heat transfer. (5)

b) Explain the concept of velocity and thermal boundary layers. (5)

c) Air at a temperature of 27°C is moving at a velocity of 0.3m/s contained in a 40W incandescent bulb. The bulb may be treated as a sphere of 50mm diameter with its surface at a temperature of 127°C. Estimate the heat transfer co-efficient and the percentage of power lost due to convection. (10)

OR

IV. a) Explain the physical mechanism of boiling. (8)

b) A flat plate 1m wide and 1m long is placed in a wind tunnel. The temperature and velocity of free stream air are 10°C and 80m/s respectively. The flow over the whole length of the plate is made turbulent with the help of a turbilizing grid placed upstream of the plate. Determine the thickness of the boundary layer at the trailing edge of the plate. Also calculate the mean value of the heat transfer co-efficient from the surface of the plate. (12)

V. a) Distinguish between: (10)

- (i) A block body and gray body
- (ii) Absorptivity and emissivity of a surface
- (iii) Total emissivity and equilibrium emissivity
- (iv) Specular and diffuse surfaces

b) A surface transmits 80 percent of the radiation incident on it in the range 0.3 to 1.0 μ and it is opaque for radiation of any other wave length. Estimate the heat flux passing through the surface if it is receiving radiation from a black surface at 1500K. (10)

OR

VI. a) Define a geometrical or shape factor. What are its salient features? (10)

b) A brick wall of emissivity 0.8, 6m wide x 4m high is located at a distance of 4m from an opening of size 20cm x 20cm in a furnace wall. The centre line of the opening lies 1m lower and 1m left of the centre of the wall. If the furnace operates at a temperature of 1523°C and the wall temperature is 37°C, calculate the radiant heat exchange between the opening and the wall. (10)

VII. a) Derive an expression for logarithmic mean temperature difference (LMTD) in the case of counter flow heat exanger. (10)

b) Water at the rate of 4 kg/s is heated from 38°C to 55°C in a shell-and-tube type heat exchanger. The water is to flow inside tubes of 2cm diameter with an average velocity of 35 cm/s. Hot water available at 95°C and the rate of 2.0 kg/s is used as the heating medium on the shell side. If the length of the tube must not be more than 2m, calculate the number of tube passes, the number of tube per pass and the length of the tubes for one pass shell, assuming $\mu_o = 1500\ W/m^2K$. (10)

OR

VIII. a) Define heat exchanger effectiveness and explain its significance. (8)

b) Hot oil having a specific heat of 2.09kj/kg K flows through a counter flow heat exchanger at the rate of 2268 kg/h with an inlet temperature of 93°C and an out let temperature of 65°C. Cold oil having specific heat of 1.67 kj/kg K flows in at a rate of 3600kg/h and leaves at 149°C. What area is required to handle this load if the overall heat transfer co-efficient based on the inside area is 0.7 kW/m^2K? (12)

IX. a) Describe the various mechanism of mass transfer. (8)

b) The air pressure in an automobile tyre reduces from 2 bar to 1.992 bar in 4 days. The volume of air in the tube is $2.5 \times 10^{-4} cm^2$, the surface area $0.5m^2$ and wall thickness 10mm. Taking the ambient temperature as 300k and the solubility of air in rubber as 3.12×10^{-3}k mol/m^3, estimate the diffusivity of air in rubber. (12)

OR

X. a) State Fick's law of diffusion. What are its limitations? (10)

b) The area of a surface of water is 0.65m x 0.65m. Air at 50°C and 1 atm with a relative humidity of 40% blows over the surface of water with a velocity of 2.3m/s. The surface temperature of water is 30^oC. Calculate the amount of water evaporated per hour per square metre of water surface. (10)

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